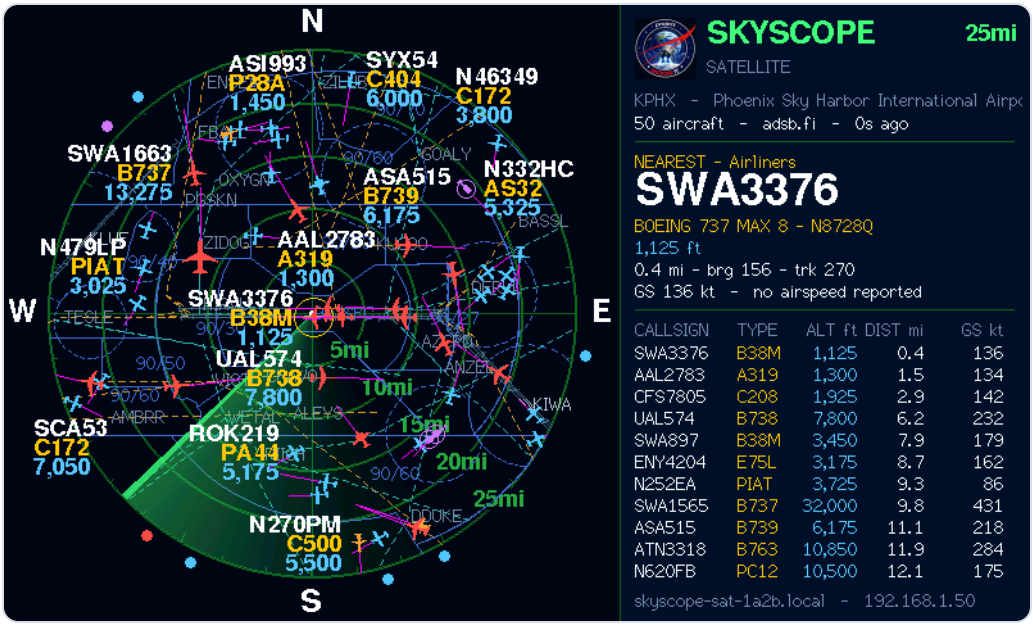
DESIGNED AND CREATED BY PROJECT MOON PI · USER GUIDE

SkyScope Satellite

A 4.3-inch display that mirrors your SkyScope radar anywhere in the house.



SkyScope Satellite: the same picture as your SkyScope, on a small screen.

Version 1.1.2 · Project: github.com/projectmoonpi/skyscope-satellite · Licence: MIT · Not for navigation.

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1. What the Satellite does

SkyScope Satellite turns a small 4.3-inch display into a second screen for SkyScope. Put it on a desk, shelf or bedside table, and it shows the same live radar as your SkyScope: the same centre and range, the same filters, the same runways, airspace and SID/STAR routes, and the same emergency and watchlist highlights.

All the work happens on the computer running SkyScope (usually a Raspberry Pi). The Satellite asks it for the picture once a second over your Wi-Fi and draws it, moving the aircraft smoothly between updates. You change everything (range, filters, map) on SkyScope's settings page and every Satellite follows. Several Satellites can follow one SkyScope.

On the Satellite	Comes from SkyScope
Radar scope with rings, sweep and aircraft symbols	Range, rings, units, sweep speed and trail, symbols and colours
Map: runways, centrelines, airspace, SIDs and STARs, labels	The Map tab's choices
Aircraft tags, the nearest-aircraft card and the list	The Filters, Presets and Alerts tabs

> **Early release.** Version 1.1.2 builds cleanly and its screen has been checked with the same drawing code running on a computer (see Building from source), but it has not yet been run on the board itself. If anything doesn't work, please open an issue with what you see.

2. What you need

Item	Details
Display board	**Sunton ESP32-8048S043**, sold as “4.3 inch ESP32-S3 display 800×480”; check the listing names the 8048S043. It has an ESP32-S3 with 16 MB of flash and 8 MB of PSRAM, an 800 × 480 screen and a touch panel. Choose the **capacitive** touch version (GT911) if offered; the resistive one works but its touch is not used.
USB cable	USB-C data cable (a charge-only cable won't let you install the firmware).
Power	Any 5 V USB supply of 1 A or more. Once installed, the computer is no longer needed.
Wi-Fi	A 2.4 GHz network (the ESP32 can't use 5 GHz).
Computer, once	To install the firmware: Chrome or Edge on Windows, macOS, Linux or ChromeOS (browser method), or PlatformIO (from source).
SkyScope	SkyScope 1.1 or later running on the same network, with its settings page on (port 7700, the default). See the SkyScope guide.
Case (optional)	A 3D-printable desk case styled like a radar console, in the case folder of the SkyScope Satellite project. It fits both the Satellite and the Mini; measure your board first, as its instructions explain.

3. Installing the firmware

There are two ways. Most people should use the browser: nothing to install on your computer.

From a web browser (recommended)

1. Download **skyscope-satellite-1.1.2.bin** from the latest release (under *Assets*).
2. Plug the display into your computer with the USB-C cable.
3. Open https://espressif.github.io/esptool-js/ in **Chrome** or **Edge**. Leave the baud rate at 921600 and press **Connect**. Pick the port that appears (often “USB JTAG/serial debug unit” or “USB Single Serial”).

   > No port listed? Hold the board's **BOOT** button, press and release **RST** (or unplug and replug the cable), then let go of BOOT and press Connect again. On Windows a driver install is occasionally needed for the CH340 chip some boards use.

4. Under *Flash Address* enter `0x0`, choose the **skyscope-satellite-1.1.2.bin** file, and press **Program**. It takes about a minute.
5. When it says it's finished, press the board's **RST** button (or unplug and replug it). The Project Moon Pi start-up screen appears.

Updating later is the same: program the newer file. Your Wi-Fi and settings are kept unless you choose *Erase Flash*.

From source with PlatformIO

For developers, or to change the code. Install PlatformIO (the VS Code extension or the command line), then:

```
git clone https://github.com/projectmoonpi/skyscope-satellite.git
cd skyscope-satellite
pio run -t upload          # builds and installs over USB
pio device monitor         # optional: the board's log, 115200 baud
```

The first build downloads the ESP32 tools and libraries (LovyanGFX, WiFiManager, ArduinoJson) and takes a few minutes.

4. First start: joining Wi-Fi

1. After the start-up screen, a new display shows **Wi-Fi setup** instructions.
2. On your phone or laptop, join the Wi-Fi network `SkyScope-Satellite-Setup` (no password).
3. A setup page opens by itself (if it doesn't, browse to `http://192.168.4.1`). Choose **Configure WiFi**, pick your network and type its password. The same page has a box for the **SkyScope Pi address**; leave it as `skyscope.local` or see the next section.
4. Press **Save**. The display joins your network and starts. It remembers the network from then on.

> To change networks later, or if the router changes, **hold the BOOT button for 3 seconds** while the Satellite is running. It forgets the Wi-Fi and opens the setup network again. Keep holding to **10 seconds** instead to restore the default display wiring (see Display hardware).

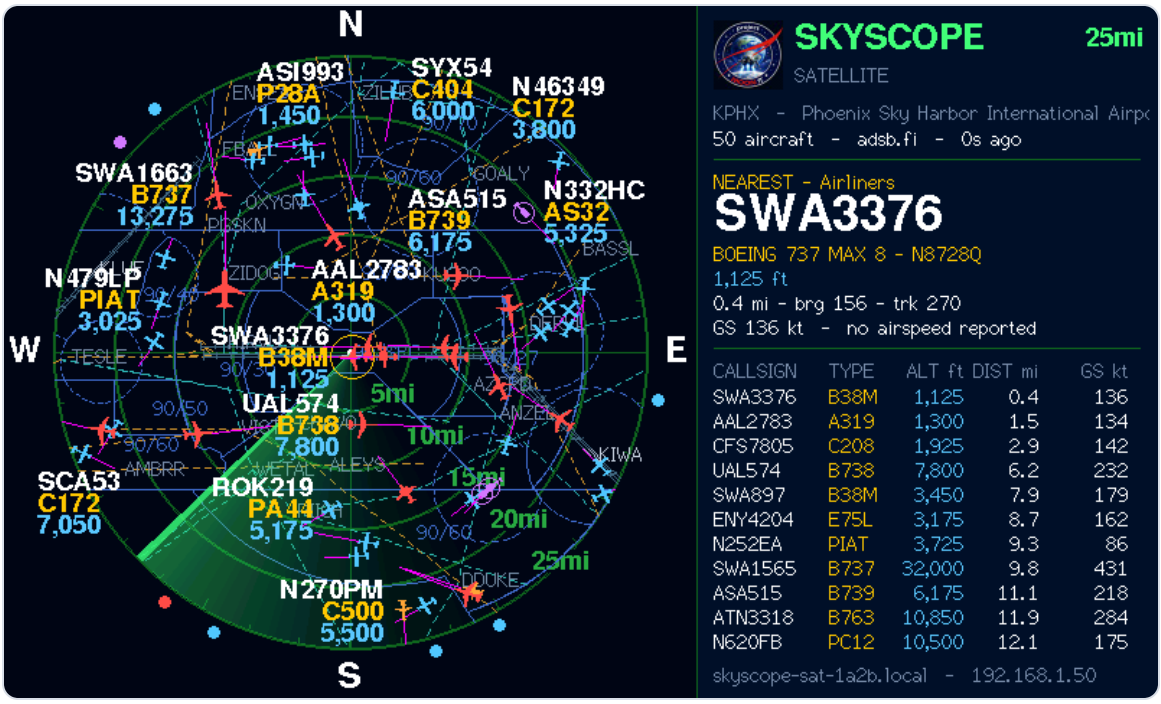
5. Connecting to your SkyScope

The Satellite looks for SkyScope at `skyscope.local`, port `7700`. There are two ways to make that work:

- **Name the SkyScope computer "skyscope".** On a Raspberry Pi: `sudo raspi-config` → *System Options* → *Hostname*, then restart it. (Raspberry Pi Imager can also set the name when you prepare the card.)
- **Or give the Satellite the address.** On SkyScope's settings page, the **About** tab shows a **Satellite address**. Type it on the Satellite's web page (section 7) or in the Wi-Fi setup page. A numeric address like `192.168.1.20` works best if your router always gives the Pi the same one (a "DHCP reservation").

When it's connected, the line under the location on the right says how many aircraft there are, where SkyScope gets its data, and how old it is. If it can't reach SkyScope, that line turns red and says why, and the Satellite keeps trying.

6. Reading the screen



Drawn by the Satellite's code from a live SkyScope feed: Phoenix Sky Harbor at 25 miles, with Class B airspace, runways and centrelines, SIDs and STARs, aircraft coloured by kind.

Part	Meaning
Rings	Distance from the centre; each ring's distance is printed where it crosses the lower-right diagonal. The outer ring is the range, shown top right.
Symbols	Aircraft, pointing where they're heading, with a line showing about a minute of travel. Silhouettes show the kind: airliner, heavy, business jet, turboprop, light aircraft, glider, helicopter.
Tags	Callsign, type and altitude. When it's busy, tags that would overlap are left out (nearest aircraft first).
Dots on the rim	Aircraft just outside the range, in their direction (if SkyScope's rim dots are on).
Yellow circle	The nearest aircraft, also shown on the card at the right.
Flashing red, pulsing ring	An emergency squawk (7500 hijack, 7600 radio failure, 7700 emergency) or emergency status.
Gold circle	An aircraft on SkyScope's watchlist.
List	Callsign, type, altitude, distance and ground speed, nearest first, alerts on top, in SkyScope's units.
Bottom right	The Satellite's own address, for its web page.

The Satellite has no touch controls: change what it shows on SkyScope's settings page.

7. The Satellite's own web page

Each Satellite has a small page of its own at the address shown at the bottom right of its screen, for example `http://skyscope-sat-1a2b.local` (the last four characters differ per board). It lets you change the SkyScope address and port, shows the connection state, and has an **About** section with the version and a link to the Display hardware page.

8. Display hardware (pins)

Boards sold as the ESP32-8048S043 have had small revisions, and similar boards use other pins. If the screen stays blank, the picture is shifted or rolls, the colours are wrong, or touch doesn't respond, you can change the display's wiring from a web page. You don't need to rebuild the firmware.

Open the **Display hardware** page in either of these ways:

- With the display on your Wi-Fi: `http://skyscope-sat-xxxx.local/hardware`, or the link at the bottom of its settings page.
- With a blank screen, or if you don't know its address: hold **BOOT for 3 seconds** and let go. That starts Wi-Fi setup. Join `SkyScope-Satellite-Setup` and press **Display hardware** on the setup page. (After saving the pins, choose *Configure WiFi* again to rejoin your network.)

Section	What's there
Panel data lines	The GPIO for each of the 16 colour bits: B0–B4, G0–G5, R0–R4. If red and blue come out swapped, swap the B and R rows.
Panel control	DE (HENABLE), VSYNC, HSYNC, PCLK and the backlight pin (-1 if there's none).
Panel timing	Pixel clock (4–20 MHz), the front and back porches and sync widths, the sync polarities, and the PCLK idle level. These come from the panel's data sheet. A picture that's shifted sideways or rolls usually needs different porches.
Touch	GT911 or none; SDA, SCL, reset and interrupt pins; and the address (0x14 or 0x5D).

Press **Save and restart**. The page checks the pins first: each must be a usable GPIO (1–21 or 38–48), and no pin can be used twice. Then the board restarts with the new wiring. **Restore defaults and restart** puts back the Sunton ESP32-8048S043 wiring, and so does holding **BOOT for 10 seconds**, even when you can't see the screen.

Default (ESP32-8048S043)	GPIO
B0–B4	8, 3, 46, 9, 1
G0–G5	5, 6, 7, 15, 16, 4
R0–R4	45, 48, 47, 21, 14
DE, VSYNC, HSYNC, PCLK	40, 41, 39, 42 (16 MHz; H porches 8/4/43, V 8/4/12; PCLK idle high)
Backlight	2
Touch (GT911)	SDA 19, SCL 20, reset 38, address 0x14

9. Troubleshooting

What you see	What to do
“Can't find skyscope.local on the network”	Name the Pi `skyscope`, or enter its numeric address on the Satellite's page (section 5). Some routers and guest networks block `.local` names between devices.
“SkyScope at … answered 404”	SkyScope is older than 1.1. Update it (`git pull` in its folder) and restart it.
Connected, but no aircraft	Check SkyScope's own radar at `http://<pi>:7700/radar`: the Satellite shows exactly what SkyScope shows, filters included.
Can't open the Satellite's page	Use the numeric address shown on its screen instead of the `.local` name.
Screen stays black, but the board's light is on	Press RST. If still black, your board revision may be wired differently: set its pins on the Display hardware page (it works without the screen). If you're stuck, open an issue with a photo of the board's back and its exact listing title.
Picture is shifted, flickers, or the colours are wrong (blue looks red)	Adjust the timing or pins on the Display hardware page. Red and blue swapped: swap the B and R rows.
Changed the pins and now nothing works	Hold BOOT for 10 seconds: the default wiring comes back and the board restarts. Or press **Restore defaults** on the Display hardware page.
“Not enough memory for the screen buffers”	The board has no PSRAM, so it is a different model. This firmware needs the 8 MB PSRAM version (ESP32-S3 N16R8).
Stuck on the Wi-Fi setup screen	Check the password, and that the network is 2.4 GHz. Join `SkyScope-Satellite-Setup` again to retry.
Want to start over	Hold BOOT for 3 seconds and let go (forgets Wi-Fi), or use *Erase Flash* in the web installer and program the file again (forgets everything).

10. Building from source and the desktop preview

The code is in C++ for the Arduino framework, built with PlatformIO. The parts: `src/feed.cpp` fetches from SkyScope on its own task; `src/feed_parse.cpp` reads its JSON; `src/render.cpp` draws the screen; `src/scope.cpp` holds the shared scope drawing (rings, symbols, tags, sweep); `src/main.cpp` starts everything and serves the web page. SkyScope's side is `/api/satellite` (traffic, once a second) and `/api/satellite/map` (map lines and labels, fetched only when they change), in ring units so the Satellite does no geography.

`include/shapes.h` holds the aircraft silhouettes and type lists. It is generated from SkyScope's own `symbols.py` by `tools/gen_shapes.py`, so all the SkyScope projects draw and classify aircraft the same way.

Desktop preview

The drawing code has no hardware calls in it, so it also builds on a computer: `tools/preview` draws one frame from saved data into an image, exactly as the board would draw it. The pictures in this guide were made that way. It needs a C++ compiler and the SDL2 development files (LovyanGFX uses them to pick its desktop mode):

```
pio run                    # once, to download the libraries
tools/preview/build.sh     # SDL_INC=… SDL_LIB=… if SDL2 isn't installed system-wide
```

The comment at the top of `tools/preview/preview.cpp` shows its arguments.

11. Credits and licence

What	Source	Terms
The radar, map and aircraft data	Your SkyScope (see its guide for its sources)	MIT
Graphics	LovyanGFX	FreeBSD licence
Wi-Fi setup	WiFiManager	MIT
JSON	ArduinoJson	MIT
Screen text	GNU FreeFont (FreeSans), as included in LovyanGFX	GPL with font exception
ESP32 support	Espressif Arduino core, built with PlatformIO	LGPL 2.1; Apache 2.0
Built with	Claude Code (Anthropic): designed, written and tested with Claude	

> **Not for navigation.** Aircraft positions may be delayed, incomplete or wrong. ADS-B shows only aircraft that broadcast it. Never use this display for flight planning or navigation.



SkyScope Satellite was designed and created by Project Moon Pi, built with Claude Code (Anthropic), and is free software under the MIT licence. Source: github.com/projectmoonpi/skyscope-satellite.